Selin **Choi**

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Education

Ph.D. Program in Culture Technology — *Daejeon, South Korea*
KAIST — *Mar. 2025 - Present*
- UVR Lab. | Advisor: Prof. Woontack Woo | AR/VR, Adaptive Playback

M.S. in Culture Technology — *Daejeon, South Korea*
KAIST — *Mar. 2023 - Feb. 2025*
- UVR Lab. | Advisor: Prof. Woontack Woo | AR/VR, Spatial Recording, Scene Graph
- Thesis: *Task-Centric Scene Graph-Based Recording with Breakpoints for Adaptive Virtual Reality Playback*

B.S. in Software, Major in System Software — *Seoul, South Korea*
Kwangwoon University — *Mar. 2018 - Feb. 2023*
- Total GPA: 4.23/4.5 | Major GPA: 4.22/4.5 | Credits: 135/133
- Undergraduate Researcher at XR Lab(2021)

Internships

VR/AR Content Lab of ETRI — *Daejeon, South Korea*
Winter Research Intern — *Jan. 2022 - Feb. 2022*
- Participated in H-BIM Intelligent Wooden Architecture Modeling Project
- Developed an AR Test Program for Traditional Buildings

Project Experience

MetaObject — *Daejeon, South Korea*
Researcher — *Apr. 2024 - Present*
- Korea Government (IITP)-funded national research project on real-time XR interface technology for environmental adaptation
- Developing adaptive guidance methods for meta-object experiences using user context, task structure, and interaction data

Dynamic Digital Twin for Realistic Untact XR Collaboration — *Daejeon, South Korea*
Researcher — *Mar. 2023 - Dec. 2023*
- Korea Government (NRF/MSIT)-funded research project on realistic XR collaboration using dynamic digital twins
- Conducted research on XR collaboration systems based on dynamic digital twin environments

AR Furniture Interior System — *Seoul, South Korea*
Team Leader — *Dec. 2021 - May. 2022*
- Developed an AR furniture placement application as a capstone design project
- Led UI development and interface design

“Dahaja” iOS Bucket List Application — *Seoul, South Korea*
Developer, Designer — *Sep. 2020 - Present*
- Independently developed and launched a bucket list application, released on the App Store in Oct. 2020
- Achieved over 65,000 downloads and managed the full software lifecycle from design to iteration

Publications

Object-Driven Event Retrieval with Assembled Object Interface for Mixed Reality Assembly Playback
2026 IEEE Transactions on Visualization and Computer Graphics (In Press)
Selin Choi, Sun Ah Bae, Dooyoung Kim and Woontack Woo

Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback
2026 IEEE Transactions on Visualization and Computer Graphics
Selin Choi, Dooyoung Kim, Taewook Ha, Seonji Kim, and Woontack Woo

Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration
2026 IEEE Transactions on Visualization and Computer Graphics
Dooyoung Kim, Seonji Kim, **Selin Choi**, and Woontack Woo

Meta-Objects: Interactive and Multisensory Virtual Objects Learned From the Real World for Use in Augmented Reality
2025 IEEE Computer Graphics and Applications
Dooyoung Kim, Taewook Ha, Jinseok Hong, Seonji Kim, **Selin Choi**, Heejeong Ko, and Woontack Woo

Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph
2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)
Taewook Ha, **Selin Choi**, Seonji Kim, Dooyoung Kim, and Woontack Woo

Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence
2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR) — Best Conference Paper Award (1st Prize)
Dooyoung Kim, Seonji Kim, **Selin Choi**, and Woontack Woo

Undo Slider for Multi-Step Editing in Collaborative AR Environments
2024 Proceedings of the Korean Institute of Information Scientists and Engineers (KIISE), pp. 31–33
Selin Choi, Minju Baeck, and Woontack Woo

Virtual Consumable Resources Interacting with Real-World Tools in a Remote XR Training System
2024 Proceedings of the Korean Institute of Information Scientists and Engineers (KIISE), pp. 248–250 — Outstanding Paper Award
Selin Choi, Yoonjae Hong, Minji Park, and Woontack Woo

Interactive 3D Voxel Drawing System Using Bare-Hand Interaction with Augmented Reality Glasses
2023 Proceedings of the Korean Institute of Information Scientists and Engineers (KIISE)
Selin Choi, Minju Baeck, Hyungwoo Jin, Taewook Ha, and Woontack Woo

Awards & Honors

Awards

2024	**Best Paper Award**, IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	*IEEE*
2024	**Outstanding Paper Award**, Korean Institute of Information Scientists and Engineers (KIISE)	*Korea*
2023	**2nd Place**, Metaverse Developer Contest	*K-META*
2022	**1st Place**, Graduation Exhibition	*Kwangwoon Univ.*
2022	**1st Place**, Startup Idea Contest	*Kwangwoon Univ.*
2021	**2nd Place**, 5G&AI-Based ICT Convergence Idea Contest	*KICS, LG U+, KT*

Honors

2023–	**Government Scholarship Recipient**, Fully funded scholarship for M.S. and Ph.D. studies	*KAIST*
2022	**Software Excellence Scholarship**, Awarded for excellence in software	*Kwangwoon Univ.*
2019	**Dean's List**, For academic excellence	*Kwangwoon Univ.*

Teaching

3D Interaction Design — *KAIST, Daejeon, South Korea*
Teaching Assistant — *Mar. 2026 - Present*
- Guiding students in Unity-based 3D interaction projects and teaching gesture recognition implementation

Augmented Reality Project — *KAIST, Daejeon, South Korea*
Teaching Assistant — *Sep. 2025 - Dec. 2025*
- Guided students in Unity-based AR project development and supported course activities

2021 Samsung Junior SW Cup — *Seoul, South Korea*
College Student Mentor — *Aug. 2021 - Sep. 2021*
- Mentored secondary school students participating in the Samsung Junior Software Cup
- Provided guidance on API usage and collaboration tools

LIKELION — *Kwangwoon Univ., Seoul, South Korea*
Instructor Assistant — *Jan. 2021 - Aug. 2021*
- Mentored beginner students in web development
- Taught Python, HTML, CSS, and basic web concepts, and supported assignments

Software Winter School — *Kwangwoon Univ., Seoul, South Korea*
Teaching Assistant — *Jan. 2021*
- Assisted freshmen in supplementary programming courses for Advanced C Programming
- Supported programming practice and answered questions about course content

Girls' Engineering Week — *WISET, Seoul, South Korea*
College Student Mentor — *Aug. 2020*
- Introduced the Software major to high school students through an online mentoring program

Extracurricular Activities

Global Shapers Community, Daejeon Hub

Daejeon, South Korea

Member

Jan. 2024 - Present

- Selected as a member of the Global Shapers Community, a World Economic Forum initiative for young leaders
- Engaging in community-driven activities through the Daejeon Hub to create positive local impact

Kwangwoon University Student Council

Seoul, South Korea

Vice President; Office Team Leader

Feb. 2019 - Oct. 2020

- Served as Vice President of the Department of Software in 2019 and organized student events, including orientations, festivals, and departmental programs
- Served as Office Team Leader of the College of Software Convergence in 2020 and managed budgets for college events

42 Seoul

Seoul, South Korea

Member

Aug. 2020 - Apr. 2021

- Completed programming challenges in POSIX-based environments using C and macOS
- Strengthened systems programming skills through implementing core library functions

Skills

Programming C#, C++, Swift, C, Python, HTML, CSS

Tools Unity, Xcode, Github, VisualStudio, Figma, Vim, AdobePhotoShop, AdobeIllustrator